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Design and Fabrication of Boost Power Factor Correction (PFC) Rectifier for Electric Scooter

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SURGE Internship Report On:
**Efficiency vs Volume tradeoff in Boost PFC
Inductor Design for Electric Scooter**

Certificate

This is to certify that the project entitled *Efficiency vs Volume Trade-off in Boost PFC Design for Electric Scooters* submitted by **Debjyoti Saha** as a part of the Students-Undergraduate Research Graduate Excellence (SURGE) 2025 offered by the Indian Institute of Technology, Kanpur, is a bonafide record of the work done by him under my guidance and supervision.

Mentor's Name: **Utsab Kundu**

Department: Electrical Engineering, IIT Kanpur

Mentor's Signature: _____

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Abstract

Power Factor Correction (PFC) is essential for electric scooter onboard chargers to ensure compliance with power quality standards and reduce total harmonic distortion (THD). This project involves the design of a Boost PFC rectifier to produce a 400V DC bus from 230V AC input, optimized for both efficiency and compactness.

A volume (LI^2)-based optimization methodology is adopted to minimize inductor size. Losses are calculated considering conduction, switching, and magnetic components. GaN FETs (Synchronous Rectification) and SiC diodes (for Diode rectifier stage) are chosen to enable high-frequency, efficient switching at 500 kHz. We attempt to figure out an optimal inductor design for our case with tradeoffs between efficiency vs volume.

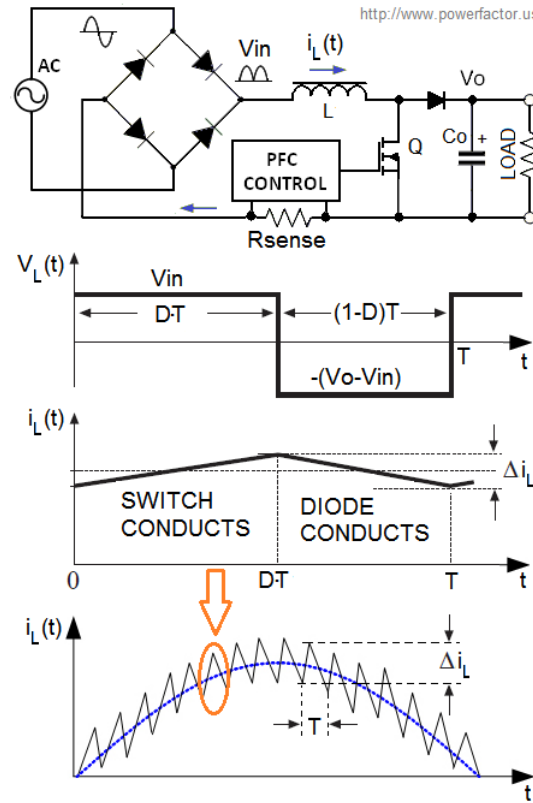


Figure 1: Boost converter with Power Factor Correction

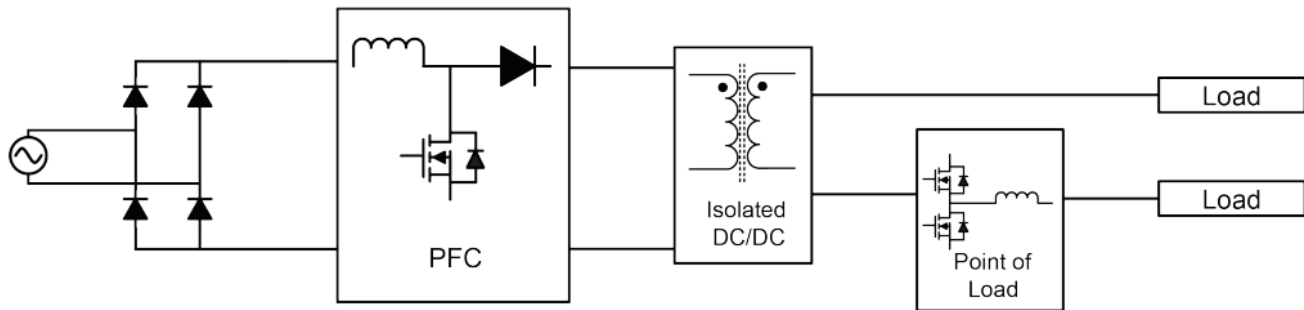


Figure 2: AC/DC Power Supply Block Diagram

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1 Introduction

With the growing demand for energy-efficient transportation, electric scooters have emerged as a popular solution. A key challenge is designing compact, efficient onboard chargers that maintain high power factor and low THD.

This project focuses on analyzing and optimizing a Boost PFC Rectifier tailored for such applications, considering component stress, loss modeling, and compliance with IEC standards.

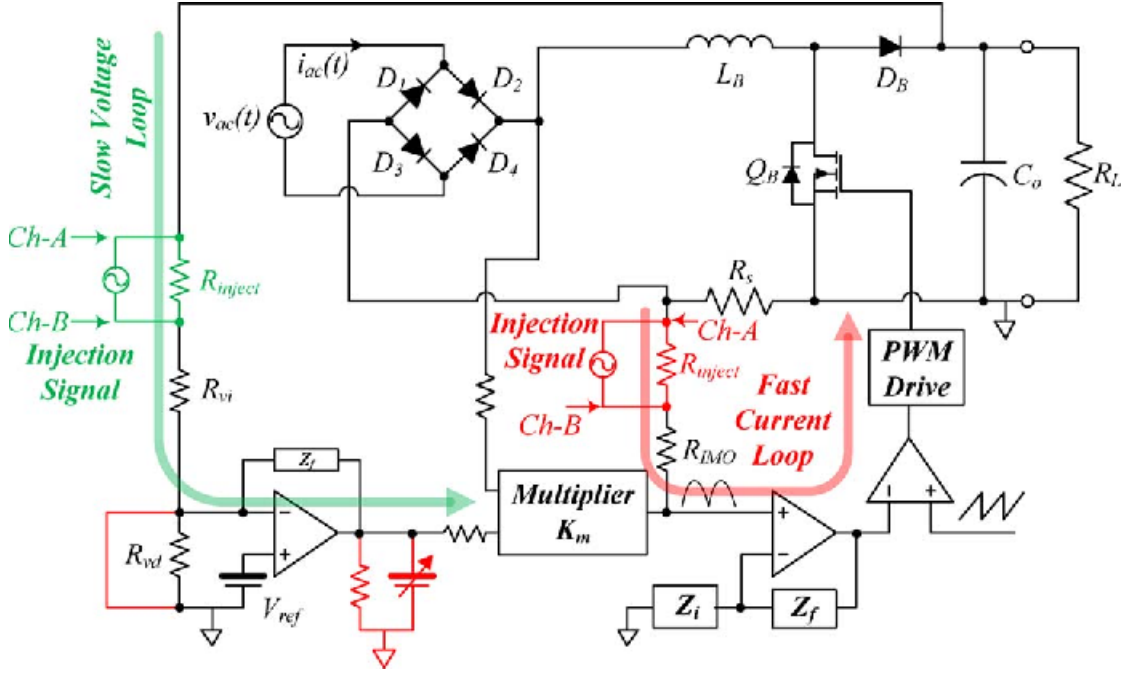


Figure 3: Typical Boost rectifier with control circuitry

2 DC-DC Boost Converter Specifications

An ideal Boost converter steps up DC voltage without losses. Practical designs include ESR, core and switching losses. The circuit operates in Continuous Conduction Mode (CCM) and is analyzed for various input voltages and load levels using MATLAB and PLECS for ideal simulation of the same.

Parameter	Value
AC Input Voltage	230 V _{rms} , 50 Hz, Single-phase
DC Output Voltage	400 V
Rated Output Power	1.2 kW
Minimum Output Power	0.24 kW
Switching Frequency	500 kHz
Input Current THD	< 5%

Table 1: Boost PFC Converter Specifications

Inductor current ripple and output voltage ripple are calculated. The switching frequency is set to 500kHz, and L, C values are derived using ripple constraints. Design equations and MATLAB plots are used.

$$\Delta I_L = \frac{V_o D(1-D)}{L f_s} \quad \Delta V_C = \frac{V_{in}}{R f_s} \cdot \frac{1}{C} \cdot \frac{D}{1-D}$$

Diode based and MOSFET based (Synchronous) rectifiers are simulated one by one and results confirm that synchronous rectification improves performance over diode-based rectifiers in terms of transient and steady-state performance.

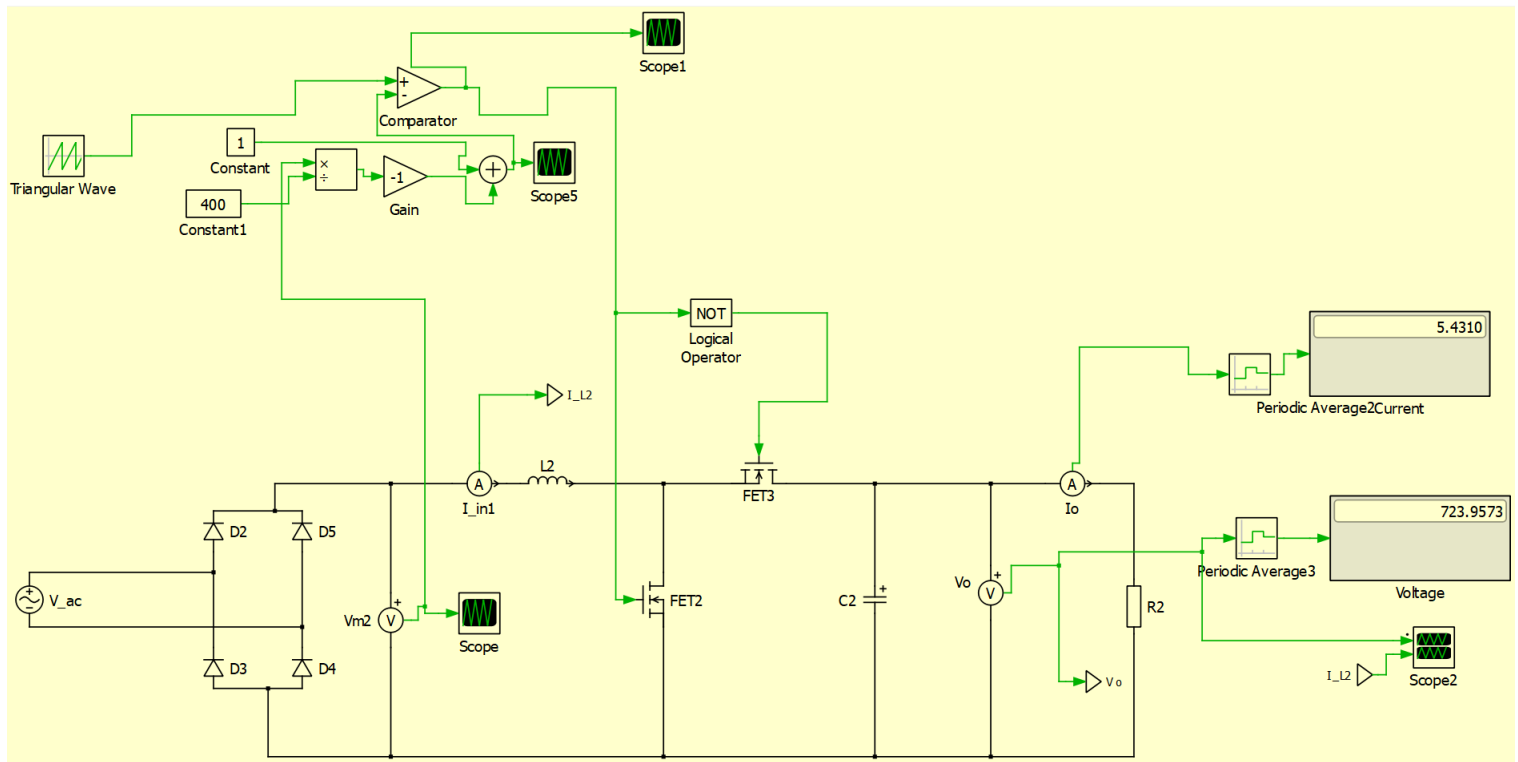


Figure 4: Open Loop model of Boost Rectifier with AC input in PLECS

A stress-index-based technique is used to select the optimum inductance value. MATLAB scripts find the minimum of maximum stress across operating points. Trade-offs between high-frequency losses and magnetic volume are analyzed. The stress index is chosen to be $LI_{pk}I_{rms}$. A grid of V_{in} and P_o is made and $LI_{pk}I_{rms}$ is found over the grid for that L value corresponding to the grid point $[V_{in}, P_o]$. Now, maxima of it is taken and the process iterates over all the points and minima of it is obtained, which gives us our desired value of L to proceed with.

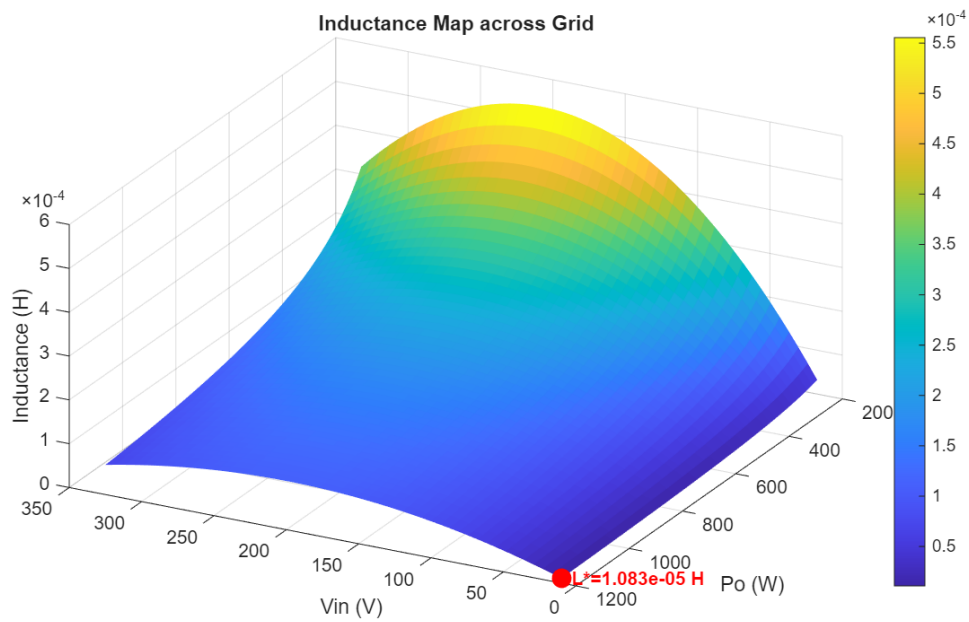


Figure 5: Plot of Inductance vs operating points

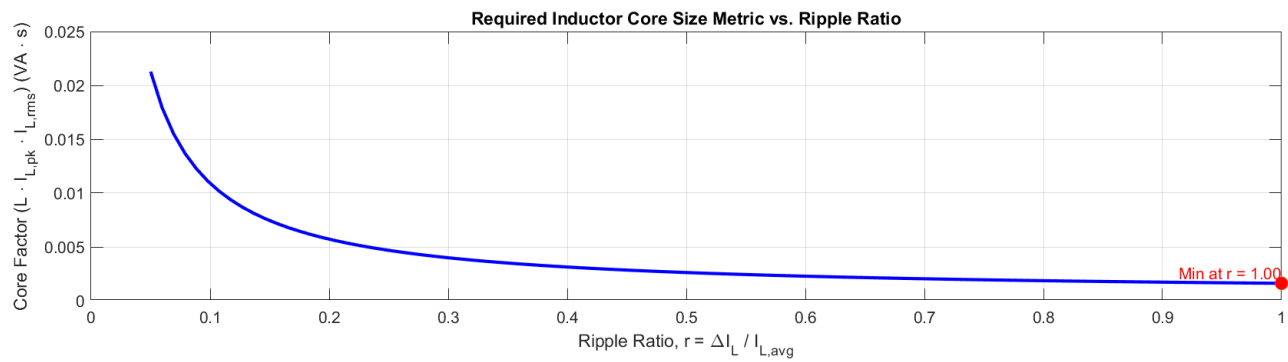
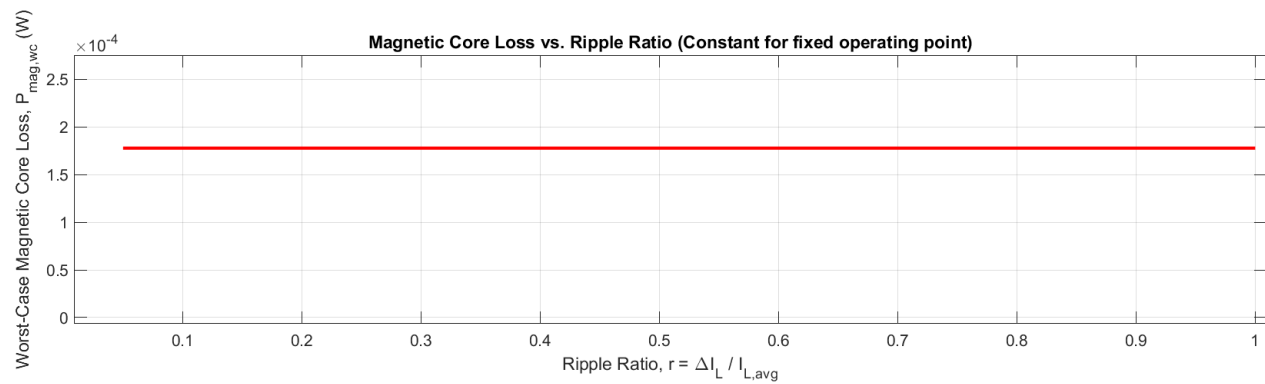
Figure 6: $LI_{pk}I_{rms}$ vs Ripple ratio (r)

Figure 7: Core loss vs ripple ratio

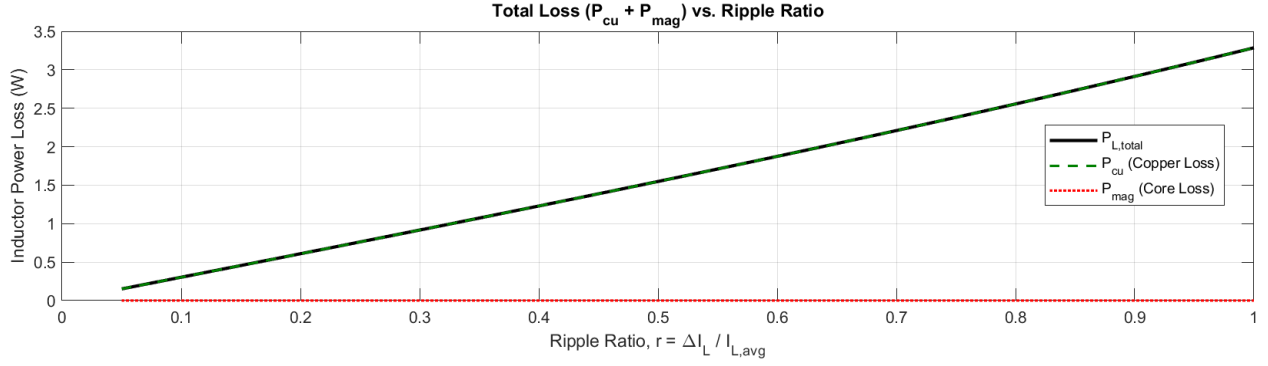


Figure 8: Worst case loss (Core + Copper) vs Ripple ratio

Optimal inductor design for the objective $J(L) = L I_{\text{pk}} I_{\text{rms}}$

Consider a boost converter in steady CCM. At a given operating point (worst case: $V_{\text{in}} = V_{\text{in,min}}$, $P = P_{\text{o,max}}$) define

$$D = 1 - \frac{V_{\text{in}}}{V_o}, \quad I_{\text{avg}} = \frac{P}{\eta V_{\text{in}}}, \quad f_s \text{ (switching frequency).}$$

The inductor ripple and currents are

$$\Delta I_L = \frac{V_{\text{in}} D}{L f_s}, \quad I_{\text{pk}} = I_{\text{avg}} + \frac{\Delta I_L}{2}, \quad I_{\text{rms}} = \sqrt{I_{\text{avg}}^2 + \frac{\Delta I_L^2}{12}}. \quad (1)$$

The ripple ratio defined as:

$$r = \frac{\Delta I_L}{I_{\text{avg}}} \implies \Delta I_L = r I_{\text{avg}}.$$

It is convenient to express L in terms of r as:

$$L = \frac{V_{\text{in}} D}{f_s I_{\text{avg}}} \implies L = \frac{K}{r}.$$

Now write the objective $J(L) = L I_{\text{pk}} I_{\text{rms}}$ as a function of r .

$$I_{\text{pk}} = I_{\text{avg}} \left(1 + \frac{r}{2}\right), \quad (2)$$

$$I_{\text{rms}} = I_{\text{avg}} \sqrt{1 + \frac{r^2}{12}}. \quad (3)$$

Therefore

$$J(r) = L I_{\text{pk}} I_{\text{rms}} = \frac{K}{r} I_{\text{avg}}^2 \left(1 + \frac{r}{2}\right) \sqrt{1 + \frac{r^2}{12}}. \quad (4)$$

Since K and I_{avg} are constants for the chosen worst-case operating point, minimizing J reduces to minimizing the shape function

$$\phi(r) = \frac{1}{r} \left(1 + \frac{r}{2}\right) \sqrt{1 + \frac{r^2}{12}}, \quad r > 0.$$

Unconstrained optimum. Differentiate $\phi(r)$. To simplify algebra, square the trivial positive factors and differentiate the monotone transformation. Direct differentiation (or symbolic algebra) yields the sign condition

$$\phi'(r) \propto r^3 - 24.$$

Setting $\phi'(r) = 0$ gives the unique positive root

$$r^* = \sqrt[3]{24} \approx 2.8845.$$

Thus the unconstrained minimum of J occurs when the ripple ratio is r^* . This value is large (well beyond typical CCM ripple limits) and therefore usually not acceptable for practical PFC/CCM designs.

Practical (ripple-constrained) optimum. In practice the ripple ratio is constrained by a specification

$$r \leq r_{\text{spec}} \quad (\text{e.g. } 0.2\text{--}0.4 \text{ for CCM}).$$

Observe that $\phi(r)$ decreases for $r < r^*$. Therefore, if $r_{\text{spec}} < r^*$, the minimum of ϕ over the feasible set $0 < r \leq r_{\text{spec}}$ is attained at the boundary $r = r_{\text{spec}}$. Substituting $L = K/r$ gives the closed-form practical design:

$$L_{\text{opt}} = \frac{K}{r_{\text{spec}}} = \frac{V_{\text{in,min}} D(V_{\text{in,min}})}{f_s r_{\text{spec}} I_{\text{avg}}} = \frac{\eta V_{\text{in,min}}^2 (1 - \frac{V_{\text{in,min}}}{V_o})}{f_s r_{\text{spec}} P_{\text{o,max}}}.$$

If the specification allowed $r_{\text{spec}} \geq r^*$, choose the unconstrained optimum

$$L^* = \frac{K}{r^*} = \frac{K}{\sqrt[3]{24}}.$$

Conclusion. The analytic minimization shows:

- The unconstrained optimum forces very large ripple $r^* \approx 2.8845$.
- Under the practical CCM constraint $r \leq r_{\text{spec}} \ll r^*$, the optimal inductance is the smallest L that satisfies the ripple constraint, i.e. $L_{\text{opt}} = K/r_{\text{spec}}$.

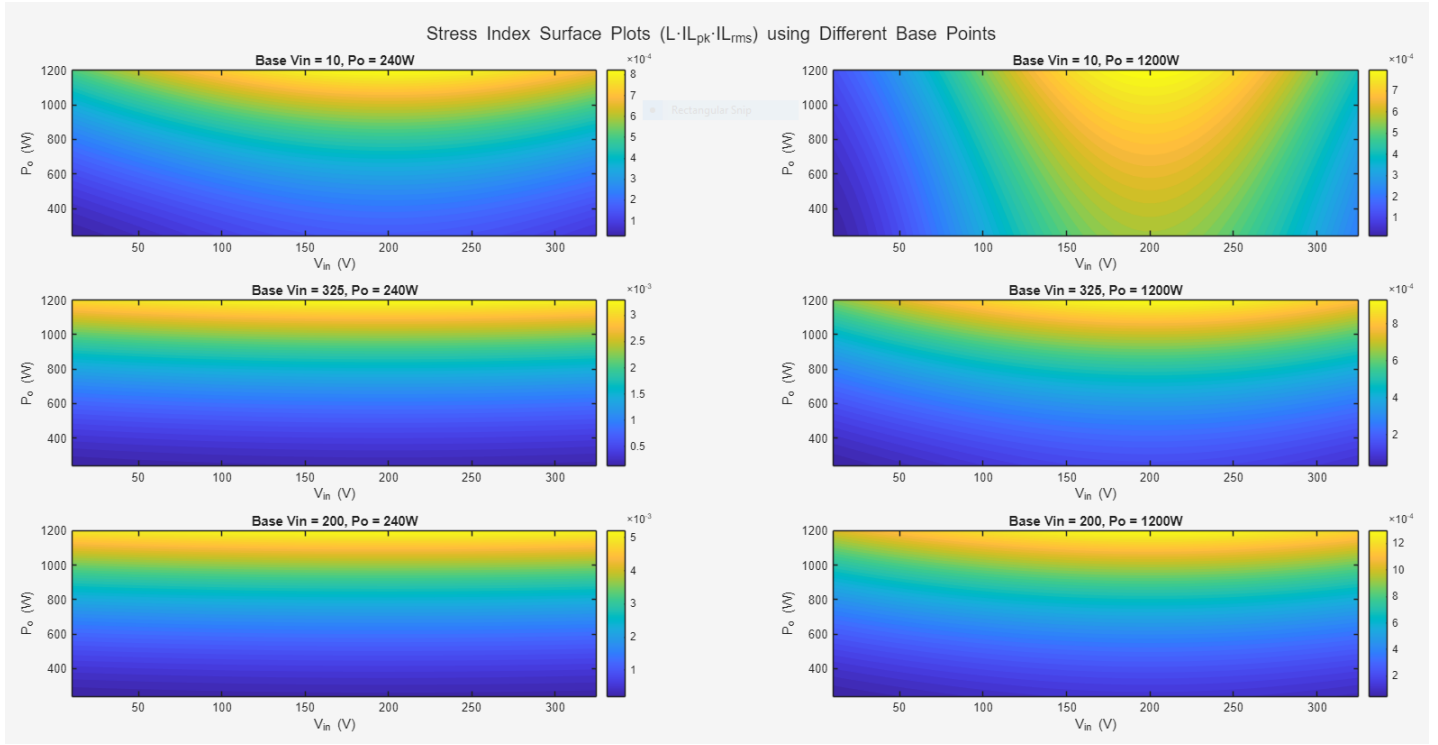
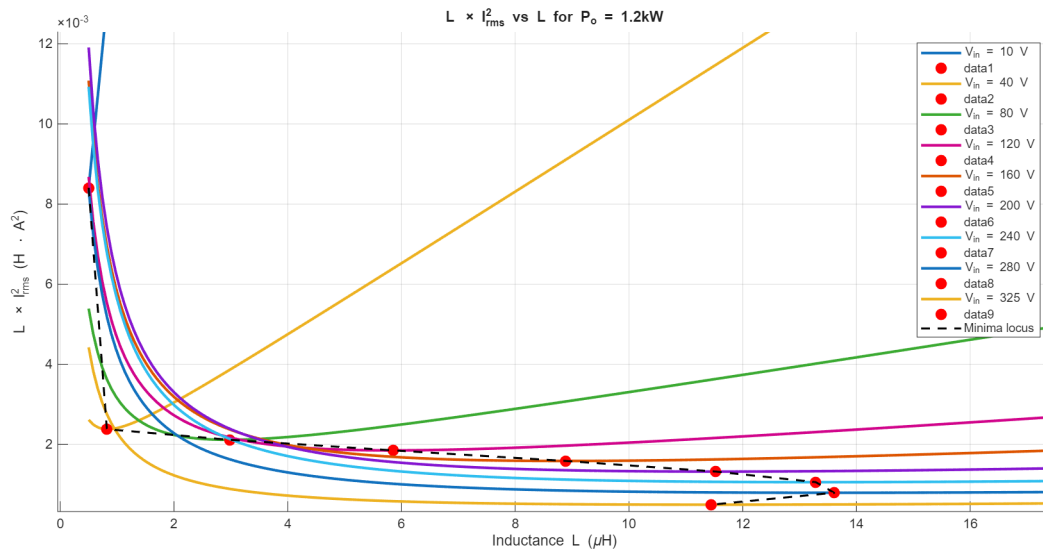


Figure 9: Stress plots for various operating points

As seen from the color plots, the $V_{in} = 10\text{V}$ and $P_o = 1.2\text{kW}$, causes the highest stress on the inductor, hence is chosen as the design point for inductor.

Figure 10: LI^2 vs L

- Objectives:

- Minimize total losses ($P_{cu} + P_{core}$)
- Minimize volume ($V_{core} + V_{cu}$)

- Constraints:
 - $B_{pk} \leq B_{sat}$, $\Delta B \leq \Delta B_{max}$
 - $J \leq J_{max}$ (copper current density)
 - Fill factor $\leq k_w$
 - Gap l_g physically realizable
- Core type: fixed (EE40 N87)
- Wire type: fixed (SWG16 litz bundle)
- Decision variables optimized by GA:
 - N — number of turns
 - r — ripple ratio ($\Delta I/I_{avg}$)

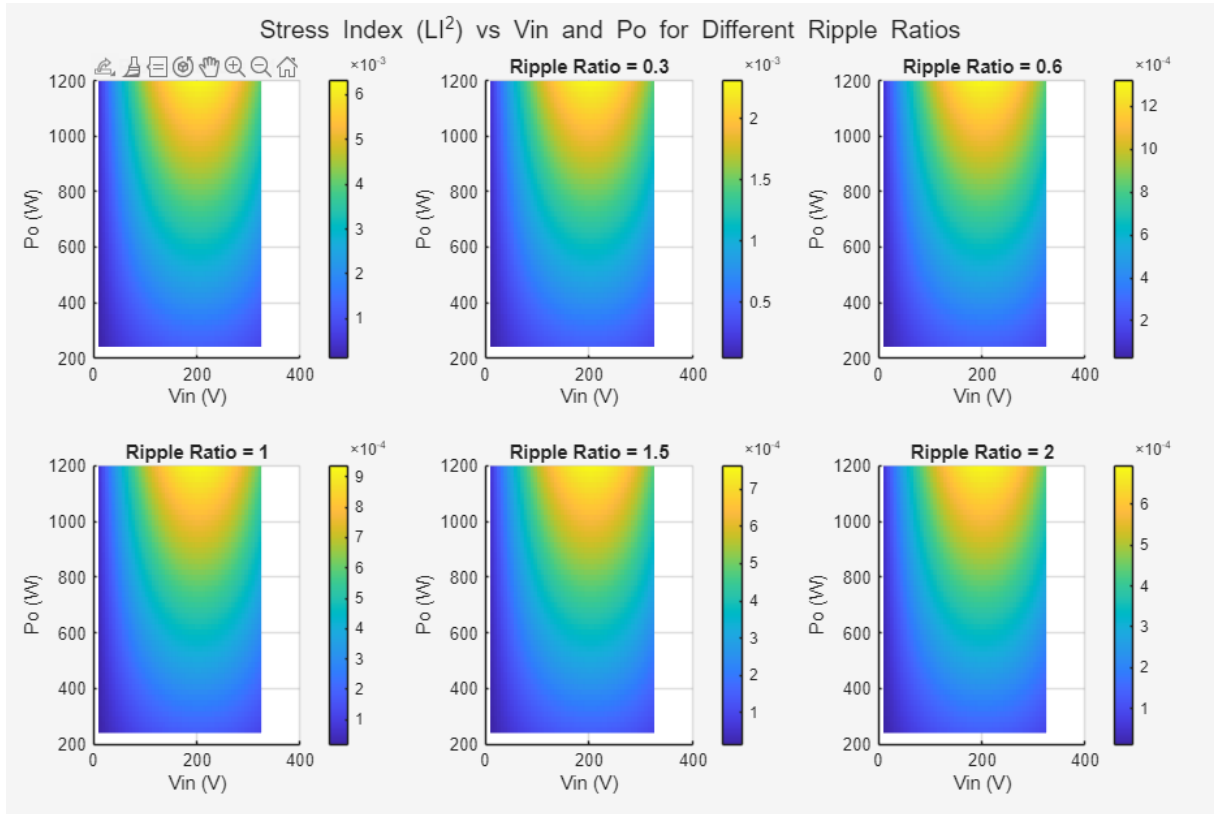
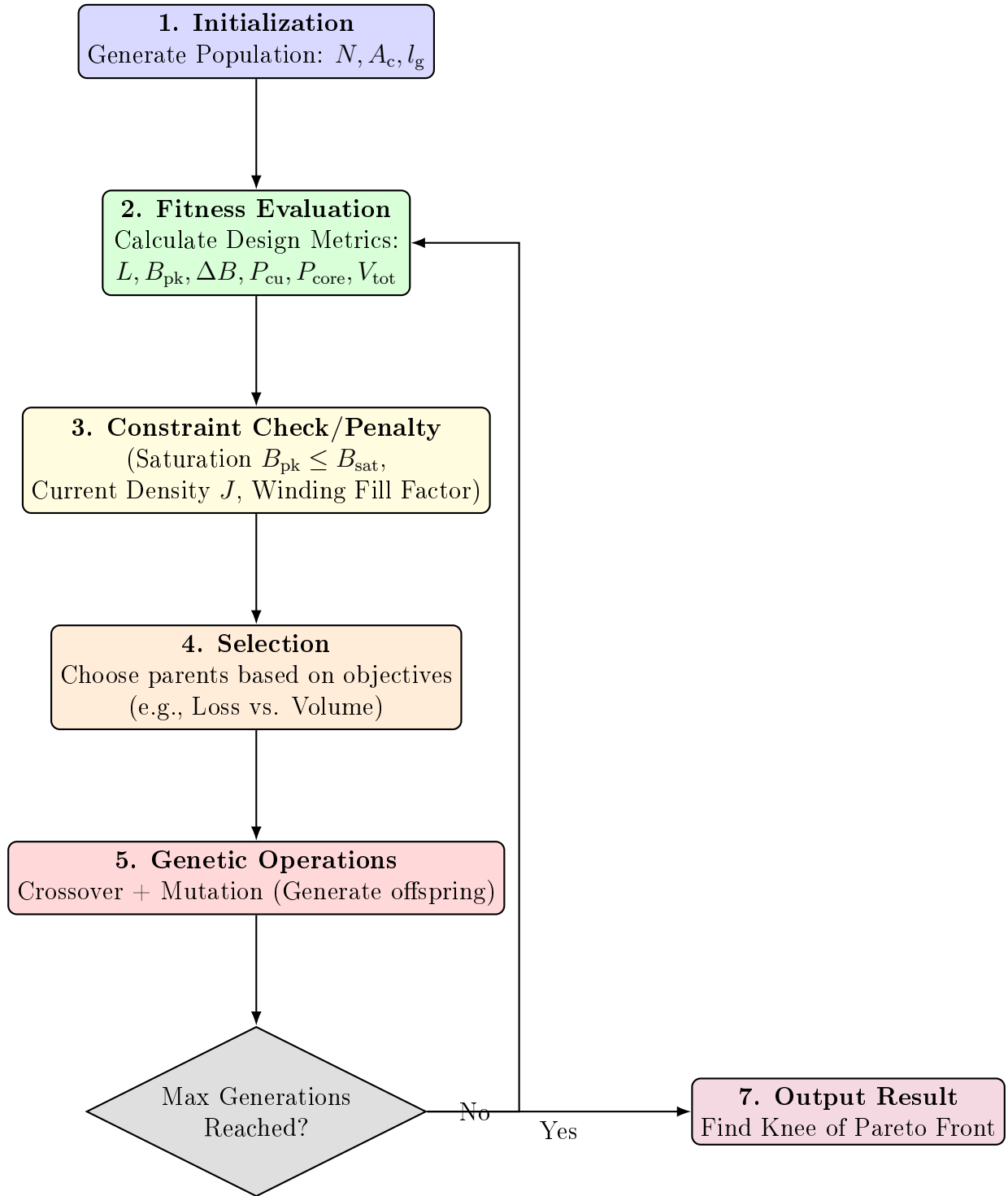


Figure 11: PCB designed in Altium



- Pareto front of:
 - Total loss vs. volume
- Knee point $\Rightarrow$ sweet spot:
 - Balanced efficiency and compactness
- Detailed design parameters:

- N, r, L, l_g
- Loss breakdown: P_{cu}, P_{core}
- Fill factor, current density J

5 Loss Analysis

Losses from conduction and core are calculated. GaN FETs (LMG3410R070) and SiC diodes are used for their low switching losses. The selected inductor is $20\mu\text{H}$, and capacitor is $3\mu\text{F}$.

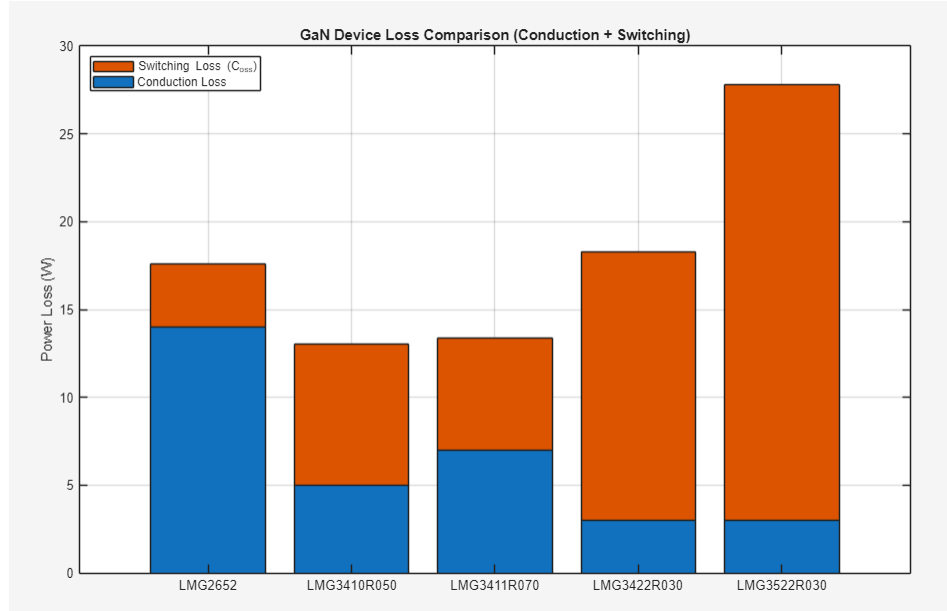


Figure 12: Comparison of available Integrated MOSFETS with drivers

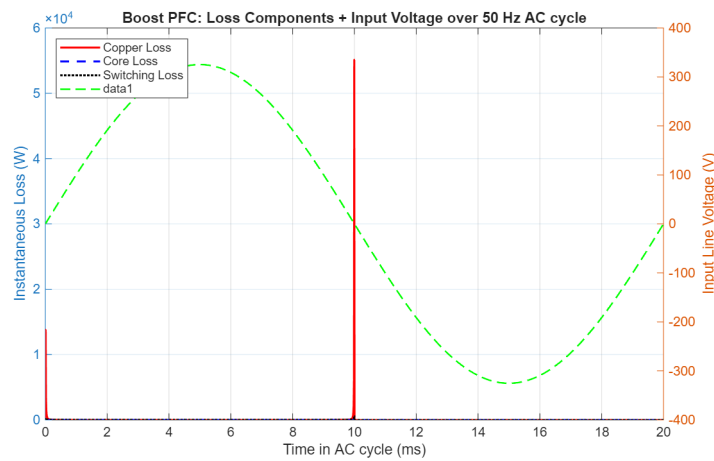


Figure 13: Loss over an AC cycle

Overall, LMG3410R050 has lowest loss but LMG3411R070 is slightly above it and has been tested on bench previously thus, it is chosen for our design.

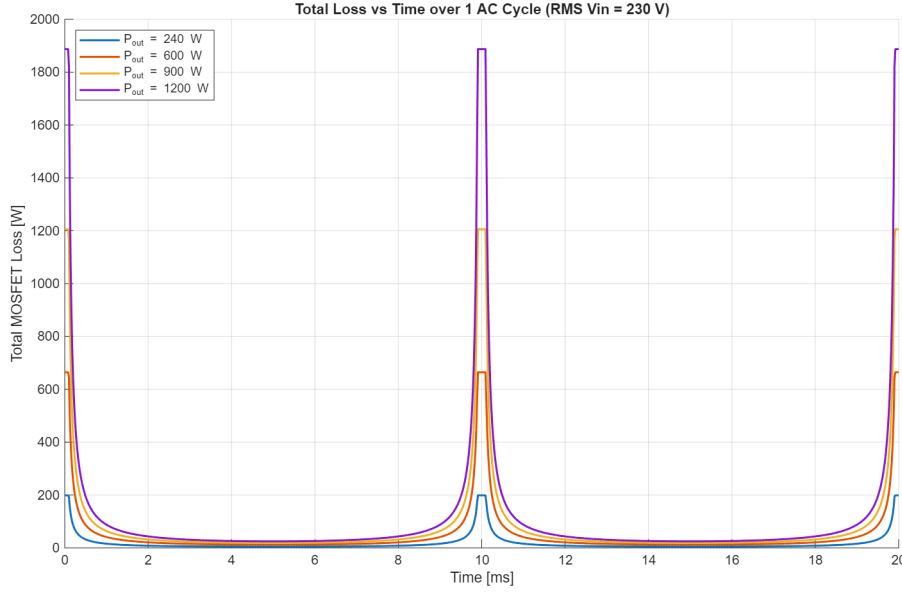


Figure 14: Total Loss over an AC cycle

6 Inductor Design Methodology and Comparative Evaluation

The inductor is the most critical passive component in a Boost PFC converter, directly affecting efficiency, power density, and thermal performance. An inappropriate choice of inductance or core geometry can lead to excessive copper loss, core loss, or impractical winding layouts. Therefore, a systematic methodology was followed to identify an optimal inductor design.

6.1 Design Framework

The required inductance is determined from the ripple-current specification:

$$L_{\text{req}} = \frac{V_{\text{in}} D}{f_s \Delta I_L}, \quad (5)$$

where $D = 1 - V_{\text{in}}/V_o$ is the duty ratio, f_s is the switching frequency, and $\Delta I_L = 2rI_{\text{avg}}$ is the current ripple expressed in terms of the ripple ratio r and average current I_{avg} .

For a given core with effective area A_e , magnetic path length l_e , and relative permeability μ_r , the required turns N and air gap l_g are obtained from:

$$N = \sqrt{\frac{L_{\text{req}} \left(l_g + \frac{l_e}{\mu_r} \right)}{\mu_0 A_e}}, \quad l_g = \frac{\mu_0 A_e N^2}{L_{\text{req}}} - \frac{l_e}{\mu_r}. \quad (6)$$

The flux swing is then:

$$\Delta B = \frac{V_{\text{in}} D}{N A_e f_s}. \quad (7)$$

6.2 Loss Modeling

Copper loss is estimated from winding resistance:

$$P_{\text{cu}} = I_{\text{rms}}^2 R_{\text{dc}}, \quad R_{\text{dc}} = \rho_{\text{Cu}} \frac{N l_{\text{mt}}}{A_{\text{cu}}}, \quad (8)$$

where A_{cu} is the effective conductor cross-sectional area, l_{mt} is the mean turn length, and ρ_{Cu} is copper resistivity.

Core loss is modeled using the generalized Steinmetz equation (iGSE) with datasheet parameters (k_i, α, β) for the chosen ferrite material:

$$P_{\text{core}} = k_i f_s^\alpha (\Delta B)^\beta V_e, \quad (9)$$

where V_e is the effective core volume.

The total inductor loss is then:

$$P_{\text{tot}} = P_{\text{cu}} + P_{\text{core}}. \quad (10)$$

6.3 Comparative Study: EE40 vs GA-Optimized Core

Two cases were studied:

1. **EE40 N87 ferrite core:** a standard catalog core with well-defined datasheet parameters (effective area $A_e = 149 \text{ mm}^2$, path length $l_e = 77 \text{ mm}$, volume $V_e = 12.1 \text{ mm}^3$, window area $A_w = 220 \text{ mm}^2$).
2. **Custom GA-optimized core:** a hypothetical core geometry evolved using a genetic algorithm (GA) with objective of minimizing total losses and volume while satisfying flux and current density constraints.

The GA formulation considered the decision vector:

$$x = [A_e, l_e, V_e, A_w, N],$$

and minimized a fitness function proportional to the weighted sum of inductor loss and volume, subject to nonlinear constraints on ΔB , B_{pk} , copper fill factor, and current density.

6.4 Results and Discussion

At the worst-case operating condition ($V_{\text{in}} = 90 \text{ V}$, $P_o = 1200 \text{ W}$, $f_s = 500 \text{ kHz}$, $r = 0.2$), the results were:

- **EE40 Core:** $P_{\text{tot}} \approx 10 \text{ W}$, dominated by copper loss, total volume 12.1 cm^3 , fill factor 0.087, current density 1.44 A/mm^2 .
- **Custom GA Core:** $P_{\text{tot}} \approx 0.09 \text{ W}$, with drastically reduced core loss due to lower flux swing, and total volume only 0.15 cm^3 .

Although the GA core suggests superior theoretical efficiency and volume reduction, such a geometry may not correspond to a manufacturable ferrite shape. In contrast, the EE40 core is commercially available and provides a feasible, validated design point. This comparison shows the value of GA-based exploration to identify Pareto-optimal designs, while practical realization is constrained by catalog availability.

7 PCB

Design

A compact 4-layer PCB is designed using Altium Designer, incorporating the chosen components and design constraints from previous analyses.

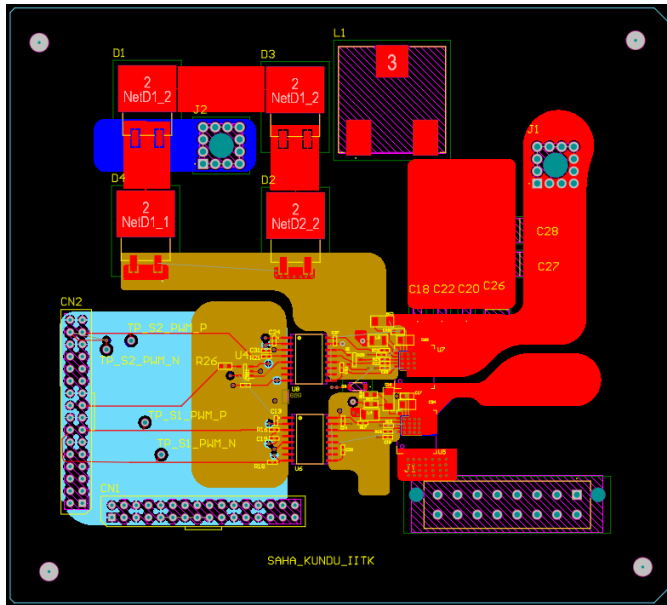


Figure 15: PCB Top Layer

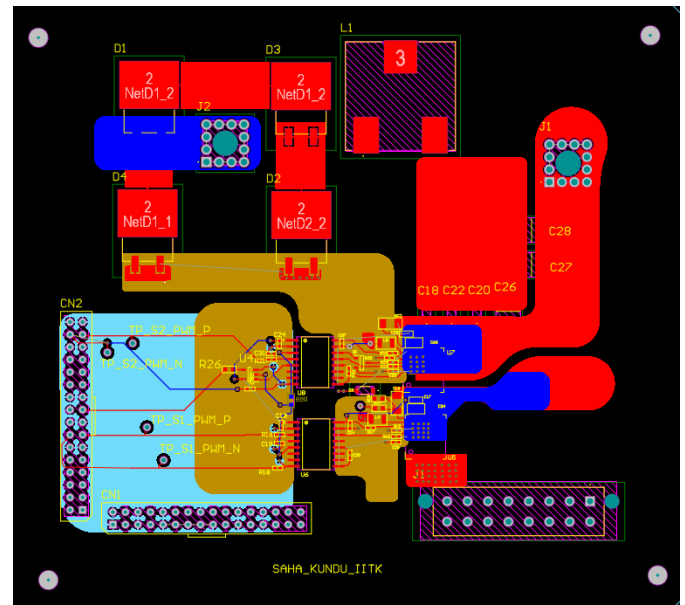


Figure 16: PCB Bottom Layer

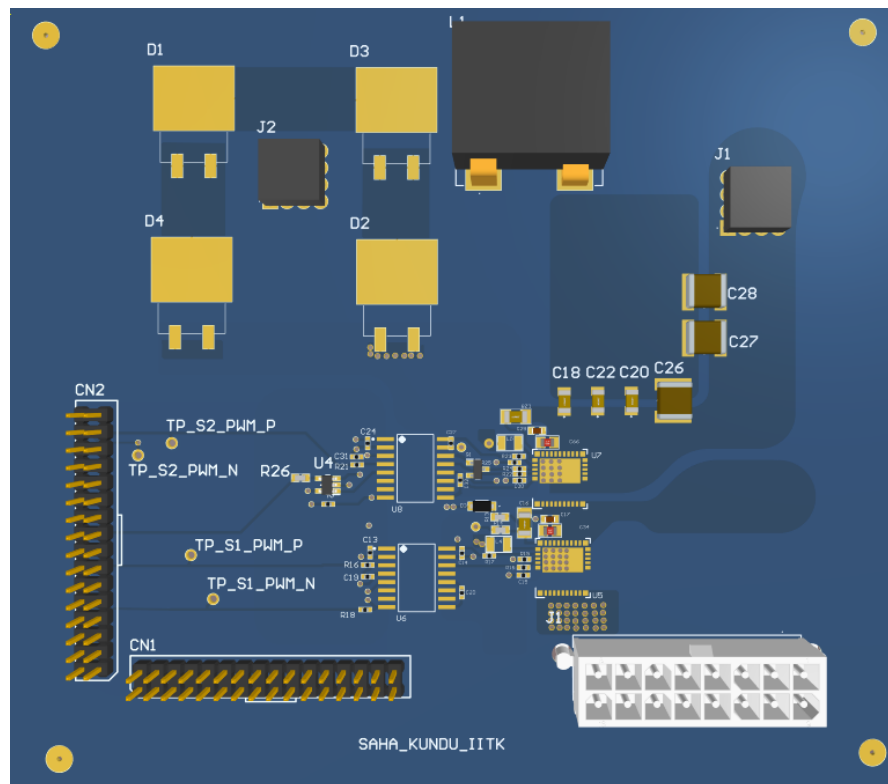


Figure 17: PCB 3D View

Component	Value/Rating	Part Number
Integrated GaN FET Switch	600 V, 70 m Ω	LMG3411R070
SiC Schottky Diode	650 V, 10 A	STPSC10H065-Y
Inductor (EE40 Ferrite Core)	20 μ H	B66381
HV SMD MLC Capacitor	0.01 μ F, 1 μ F	2220CC105KAZ2A, C1206C105K1RAC

Table 2: Key Components and Specifications

8 Conclusion

An inductor value of around 22 μ H provides an optimal balance between efficiency and volume, achieving losses below 0.35%. This trade-off is essential for practical EV charger design.

9 Future Work

- Prototype testing
- Use of advanced magnetic materials
- EMI-aware and thermally-optimized design
- ML-based adaptive design automation

10 References

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